

Total No. of Questions : 8]

SEAT No. :

PD4209

[Total No. of Pages : 2

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T.E. (Civil Engineering)

WATER SUPPLY ENGINEERING

(2019 Pattern) (Semester - V) (301002)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Figures to the right side indicate full marks.
- 3) Use of calculator is allowed.
- 4) Assume Suitable data if necessary.
- 5) Neat diagrams must be drawn wherever necessary.

- Q1) a)** Explain in detail: Rapid Mixer and flocculator. **[6]**
b) A clariflocculator is to be designed for a flow of 60 MLD. Assuming, inlet velocity 1.2 m/s, and detention time in flocculator and clarifier as 30 and 120 mins, respectively, surface overflow rate in clarifier as $40 \text{ m}^3/\text{m}^2/\text{d}$, G in flocculator 40 s^{-1} , and depth of flocculator basin as 2.5 m, determine diameter of inlet pipe, flocculator and clarifier. **[6]**
c) Discuss in detail filtration process in Rapid Sand Gravity filter with neat sketch. **[4+2=6]**

OR

- Q2) a)** Explain in detail diffused double layer theory of coagulation with neat sketch. **[4+2=6]**
b) A filter unit is of size $5\text{m} \times 10 \text{ m}$. After filtering $9000 \text{ m}^3/\text{day}$ in 24 hr period, the filter is back washed for 30 minutes at the rate of $10 \text{ lit}/\text{m}^2/\text{sec}$ for 10 minutes. Compute the average filtration rate, quantity and percentage of treated water used in washing and the rate of wash water flow in each trough. The unit has 4 collecting troughs. **[6]**
c) Discuss in detail coagulation by natural coagulant *Moringa oleifera* stating its advantages and disadvantages over Alum. **[6]**

- Q3) a)** Explain in detail following water softening processes: **[6]**
i) Lime soda process
ii) Zeolite process
b) What is Super Chlorination? Find the dose of chlorine and chlorine demand for water quantity of $50,000 \text{ m}^3/\text{day}$. Chlorine used is 20 kg per day and residual chlorine after 10 minutes of contact time is $0.2 \text{ mg}/\text{lit}$. **[6]**
c) What do you mean by disinfection? Discuss the factors affecting efficiency of disinfection. Enlist 4 disinfectants used in water treatment plant. **[1+2+2=5]**

OR

P.T.O.

- Q4) a)** Explain in detail: various methods of desalination of water with neat sketch. [6]
 b) Explain in detail: Break Point chlorination with neat sketch. [4+2=6]
 c) Write with a neat sketch: a detailed note on Ion exchange. [2+3=5]

- Q5) a)** Write in detail: Methods and Components of domestic rain water harvesting with a suitable sketch. [8]
 b) Designed demand of town is 4 MLD. It is pumped into an elevated service reservoir at a uniform rate from 5am to 9am and 5pm to 9pm. The variation in demand of water is given below. [10]

Period	5 am to 9 am	9 am to 5 pm	5 pm to 9 pm	9 pm to 12 am	12 am to 5 am
Demand	40%	15%	30%	10%	05%

Determine the balancing Capacity of the reservoir.

OR

- Q6) a)** Explain in detail: Requirements of good water distribution system. [6]
 b) Calculate storage capacity and dimensions of the tank to store rain water for the given data: [6]
 Terrace area=300 m², average annual rainfall =720 mm
 Runoff coefficient= 0.8, Assume L/B= 2, D=2m
 c) List down various leakage detection systems. Write a detailed note on any one leakage detection system. [2+4=6]

- Q7) a)** Explain in detail with neat sketch: Packaged WTP for swimming pools. [2+4=6]
 b) Write a note on: [6]
 i) Air valve,
 ii) Sluice valves
 c) Explain with neat sketch two pipe system. [5]

OR

- Q8) a)** Explain following initiatives taken by Government in water infrastructure. [3+3=6]
 i) Amrut mission
 ii) Jal Jeevan mission
 b) Explain in detail requirement of water meters and state their types. [3+2=5]
 c) Write a note on: [6]
 i) Drain valve,
 ii) Vent pipe

